



Dear Study Members,

Welcome to our Autumn Newsletter!

This month we are celebrating the **20th anniversary** of the official launch of the Breast Cancer Now Generations Study. We want to take this opportunity to **thank you again for your participation and continued support** of the study over the past 20 years.

In this issue, you can see the amazing amount of data that has been collected, some of the important scientific findings and the key questions about breast cancer that we are currently asking. We hear from Dawn Thomas, our study manager, and Michael Jones, our statistician, about their memories of the launch of the study. We're also delighted to share reflections from two of the first participants to join Generations, Jan Brooks MBE and Amanda Jones.

Last year, the team at The Institute of Cancer Research, London, developed plans for the next five years of the study and we are very happy to report that Breast Cancer Now has approved the funding for them.

One of the first activities in this five-year plan is the next questionnaire, which you will be receiving soon. We have created this questionnaire so it can be completed easily on your phone, tablet or computer. It's important that we keep getting a high response rate to the questionnaire to maintain the high quality of our research so thank you for your continued involvement. And, for the first time, we will be sending invitations and reminders by email and text message. **To help us keep in touch, please let us know if you change your email address or mobile phone number [by completing this form](#).**

We look forward to working with you over the next five years, and beyond!

With best wishes,

Amy Berrington Professor Amy Berrington

M García-Closas Professor Montserrat García-Closas

A 20-year journey: Highlights from the Generations Study

★ High-Quality and Extensive Data Collection

- **Over 70% of participants have completed 5 questionnaires**
- **Blood samples from more than 100,000 participants**
- **NHS breast screening records for over 70,000 participants**
- **Mammograms from over 4000 participants**
- **12,000 women with physical activity tracker data**
- **Breast cancer tissues samples from over 2000 participants**

★ Key Scientific Findings:

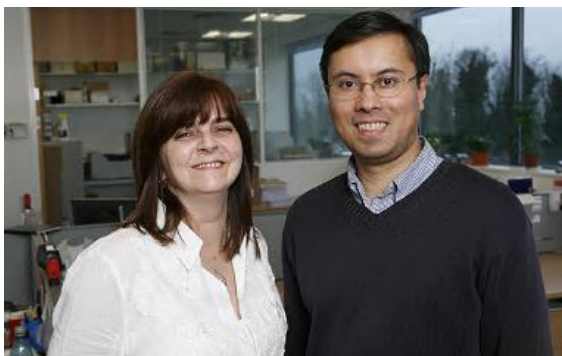
- **Discovery of genetic risk factors:** A newly developed genetic risk score can help to identify high-risk women, even if they do not have relatives with cancer. This could help with personalised prevention (for example, by taking risk-reducing drugs) and early detection (such as more frequent screening) for women at high risk.
- **Reproductive history and cancer risk:** 10 years after having a baby, women are less likely to get breast cancer compared to those who never had a baby. Although it is uncommon to have cancer soon after having a baby, the risk for having an aggressive cancer type (triple-negative) is slightly increased.
- **Body weight and breast cancer risk:** Higher body weight after menopause increases breast cancer risk. However, before menopause, higher body weight is linked to a lower risk of developing early onset breast cancer, and this connection is stronger than previously thought.
- **Physical activity and cancer risk:** Engaging in more leisure time physical activity can lower the risk of breast cancer before menopause.

★ Some priorities for the next five years building on our progress so far:

- **Better breast cancer risk tools:** Improve tools for predicting breast cancer risk by adding the latest discoveries such as genetic scores, artificial intelligence (AI) analyses of mammograms, and social and environmental factors. This will help women receive more personalised and effective care.
- **Lifestyle changes and breast cancer prevention:** Investigate how changes in lifestyle such as increasing physical activity can reduce the risk of breast cancer and improve survival after a breast cancer diagnosis. This will help women make informed choices to protect their health.

- **AI analyses of breast tissue:** Use AI to analyse breast tissue samples to help us find out what types of benign breast disease, such as fibroadenomas, are linked to a higher risk of developing breast cancer. This will help women with these benign conditions to understand their cancer risk and take preventive measures.
 - **Securely share Generations Study data:** Make the Generations Study data securely available to the wider research community to enhance breast cancer research. This will accelerate discoveries and improve outcomes for women.
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Dawn and Michael look back on the launch of the study



Dawn: I was here at the launch of the study 20 years ago. I'd been brought in to oversee the team that was going to manage the recruitment and communications with participants.

Michael: I remember the launch well. We were unsure about how many women were going to join the study at the start. We had been running a pilot study to get our documents and procedures right, but this was only with a relatively small number of women.

Dawn: Breakthrough (now known as Breast Cancer Now) did an exceptionally good job with publicity for the launch. We featured on the [BBC website](#), in national newspapers, and Prince Charles (now His Majesty the King) supported the study in an article in the Sunday Express. We became famous overnight.

Michael: On one hand we were afraid that not enough women would be interested, and on the other, we were worried about being swamped with requests.

Dawn: Yes, it was all quite daunting not knowing what to expect but we didn't have to worry as we had a huge response. On launch day, we had over 9,000 women signing up on our

website, and 6,000 women leaving voicemail messages. It was a case of 'all hands on deck' to deal with this during the first few weeks.

Michael: A few months after the launch it became less frantic. Dawn's mailing operation was a smooth piece of work.

Dawn: The logistics were often challenging, as each week we were posting thousands of invitations to women, logging their replies, and sending out questionnaires and blood packs. We got to know the Royal Mail very well as we filled the collection vans with sacks full of post each day and received sacks full of responses.

Michael: We were very excited when we recruited the first 1,000 participants, and we had a small celebration. We did that at 5,000 recruits as well, and by 10,000 we knew we could reach our target.

Dawn: It was challenging work, but all extremely exciting and a deep sense of achievement as we hit new targets and celebrated each with a cheeky tipple. Over the years, I have had a wonderful team working with me with great camaraderie and unwavering support. We've all felt it was a worthwhile cause and 20 years on, we are proud of our achievements. None of this would have been possible without the support from Breast Cancer Now and the dedication and motivation of you, the participants in the study.

Michael: Thank you for your valued contributions to our research and your ongoing support and participation, we are extremely grateful for your time and commitment.

Meet fellow participants and hear about their experiences

Jan Brooks MBE



Before the Generations Study was set up, I was a very active supporter of Breakthrough Breast Cancer (now known as Breast Cancer Now), which I joined after I was diagnosed with breast cancer in 1992. I joined this organisation as I was passionate about breast cancer research and wanted to find out as much as I could about it.

When Professor Alan Ashworth and Professor Tony Swerdlow from The Institute of Cancer Research presented their ideas to find out the causes of breast cancer in a long-term study spanning 40 years, Breakthrough Breast Cancer was very supportive. They set up an advisory panel for the research and I was a participant representative on it. I was delighted that we were doing such a study, which would be different to other previous studies because it involved giving a blood sample to the researchers and would therefore be able to give them much more knowledge to go on than simply a survey. I continue to participate as I believe in the study and the important results over the last 20 years.

I have found it easy to participate and I have encouraged many others on the Isle of Man where I used to live, to join as well. My daughter, who is now a doctor herself, joined with me. When we were fundraising for breast cancer research and giving presentations, I found that the Generations Study was a very good subject to talk about as people are genuinely interested in the causes of breast cancer.

My husband died a few years ago, so my family is now my daughter, her husband, my twin granddaughters, and my son and his partner. I am retired from teaching biology in a secondary school, and my hobbies are music, singing and walking.

My diagnosis was a late one because I had presented nearly a year earlier with a lump in my breast but was told by the general surgeon that it would be benign, as with one or two others I had. After 10 months, I became increasingly worried and asked if I could have it removed, and a biopsy done. This was done and it was diagnosed as grade 2 oestrogen receptor positive breast cancer, which involved my lymph nodes. I had surgery, chemotherapy, radiotherapy and tamoxifen.

My life carried on as normal but always with the worry of the breast cancer returning. In 2005, I found I had another lump. This time it was diagnosed quickly, and I had a mastectomy and reconstruction. It was just grade one, so I didn't have to have any treatment.

In 2017, I had extreme pain in my left arm, which has resulted in paralysis of my left hand from original radiotherapy damage in the brachial plexus, and I was diagnosed with tumours in that area as well as other areas. I have had very good treatment since then with palbociclib, a new targeted drug, which has kept the metastasis pretty much under control. However, I now have some small lung metastasis but I am told that I will receive another targeted hormonal drug.

I would like to say that my positive attitude is due to the advances in breast cancer research and, when the Generations Study started, I thought this was so important for future generations to understand more fully the causes of breast cancer. I am very grateful to everyone who works on the Generations Study.

Amanda Jones



In 2003, my beautiful and beloved youngest child, Rebecca, died of breast cancer at the age of 32, just 11 months after being diagnosed. To face living without her, I contacted Breakthrough Breast Cancer, and, in desperation, I asked what I could do to help make breast cancer different for future generations -- both in terms of research and its outcomes. Through this original contact, I learnt about the Generations Study, and all that it meant, what it aimed to do, and how it was going to be a permanent part of exploration into breast cancer from 2004.

I signed up at once, along with my daughter Camilla, and my daughter-in-law Joanna. It was easy for us all and we knew that we were truly helping to make a big step into discovering the problems and varieties and dangers of breast cancer at that time. Having done this, I knew that I had committed to what was long-term, ongoing, developing exploration of breast cancer and I wanted to be part of this road of discovery for my darling Becs and every future person, woman or man, who developed breast cancer.

The process of participating has always been very uncomplicated – straightforward to complete and always received with gratitude and encouragement.

Born in 1942, I was educated at boarding schools. My father spent the war in Burma, and I didn't see him until I was over three years old, when he returned, thankfully uninjured, to our flat in London. I was an only child, but my parents did everything to give me a varied and very happy childhood.

I studied at Westminster Hospital and graduated in radiography, a qualification which took me to London hospitals, hospitals in the Highlands of Scotland and in the south of France.

I married in 1964 and had three children, Camilla, Dominic and the youngest, Rebecca. We had (adored) cats and dogs of varied breeds, had family holidays in this country and abroad when possible. I enjoy being a member of book groups, French-speaking groups, swimming and being with my grandsons. A very blessed time. I took to running, an activity which gave me many chances to raise money for Breakthrough Breast Cancer and Breast Cancer Now. Over the years, I have raised a total of £300,000, to help with other projects at Breast Cancer Now.

Our family's experience of breast cancer completely shattered our life in 2002, when Becs rang me from her home in France to say she had found a lump in her breast and her husband was going with her to the doctor that afternoon. Life changed for us all. Becs was immediately admitted to hospital to have the tumour removed and we all arrived in France to be with them both.

From then until she died, our families constantly commuted to and from France as her treatments accelerated. She had a mastectomy after two months and we all went up to the mountains to have New Year together, with our beautiful girl, with not a hair on her head, dancing on the table. We all felt New Year was making a difference, along with Herceptin and Taxol. But then in March, we all left for France as secondaries had spread uncontained and

there was nothing else medically to be done beyond palliative care and we were with her when she died. She was in no pain and looking as beautiful as she always was.

I want to express just how much I, Camilla and Joanna, and their husbands, want us to always continue in the Generations Study, to participate in the advancements and the progress, both in memory of our beloved Becs and in celebration of all those who will be saved from breast cancer, and its effects, by the scientific successes resulting from the Generations Study.



Camilla



Joanna